

Pietro Maria Sparago

GENERAL INFORMATION

E-mail: p.sparago@lse.ac.uk (LSE) / pietrospar@gmail.com (main)
Personal webpage: <https://pitspa.github.io>
Google Scholar: Pietro Maria Sparago
LinkedIn: Pietro Maria Sparago
Mathematics SE: Snoop (top contributor; ~20k reputation)
LeetCode: pitspa
GitHub: pitspa

EDUCATION

LSE, London, UK

September, 2022 - Current

PhD, Statistics. Expected thesis submission: Q3 2026.

- Obtained the highly competitive LSE PhD Scholarship.
- Research group: *Probability for Finance and Insurance*. Research in mathematical finance, stochastic processes and stochastic analysis. Applications in numerical simulation schemes of diffusions under exotic terminal conditions (e.g. stopping, barrier options).
- Presented at the World Bachelier Congress 2024 (Event sponsors: G-Research, SIG Susquehanna, Ripple).
- MSc GTA: Bayesian machine learning (ST451). Discrete-time state space modelling (dynamic linear models, dynamic Alpha-Beta, Kalman filter), Bayesian regression and classification, introduction to Markov chain Monte Carlo (posterior Gibbs sampler).
- BSc GTA: stochastic processes (ST302), quantitative finance (ST330) and mathematical statistics (ST202).
- Leadership role: PhD Students' Representative. Voiced my cohort's requests and issues to senior academic staff, proposed + negotiated actionable solutions regarding funding and complex IT requirements (e.g. internal computing clusters).
- Advisors: Prof. Umut Çetin, Prof. Angelos Dassios.

Bocconi, Milan, Italy

MSc, BSc, Finance (110/110 *Cum Laude*, GPA 29.43/30 (98%)).

EXPERIENCE

Bank of England, London, UK

August, 2023 - February, 2025

PhD Researcher, Interest Rates and Asset Pricing team - Internship

- Worked on proprietary options and futures markets data: built and maintained a custom volatility surface engine in Python for online daily raw option price data; static replication and testing code for relevant market-implied signals.
- Participated in preliminary Monetary Policy Committee meetings.
- Work published on the Bank Overground 2025.

CODING SKILLS

Python (preferred): `numpy`, `scipy`, `pandas`, `pytorch`, `scikit-learn`, `numba`.
Expert use of AI for coding and agentic workflows (GPT 5.5, Opus 4.8) (visit my GitHub!).
Also: experience with R, SQL, C++, Matlab. Highly proficient in $\text{\LaTeX}$.

LANGUAGES

English, Italian, Spanish B1